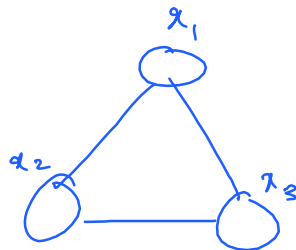


Chapter 13

Factor Graphs and Sum/Max-product Algorithms **

This MRF implies the factorization



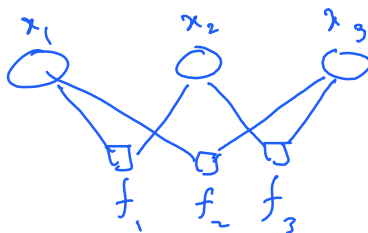
$$p(x_1, x_2, x_3) \propto \psi(x_1, x_2, x_3)$$

But suppose we actually want to represent

$$p(x_1, x_2, x_3) \propto f(x_1, x_2) f(x_2, x_3) f(x_3, x_1)$$

Is there a way to do this with a graph?

Factor graph:



Two types of nodes:

V : variables: x_1, x_2, x_3

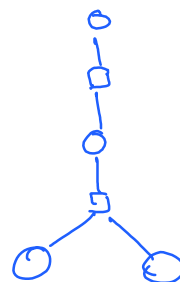
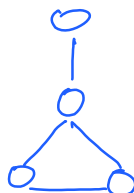
F : factors: f_1, f_2, f_3

$$p(x_1^m) = \prod_{f_i \in F} f_i(x_{f_i})$$

variable nodes adjacent to f_i

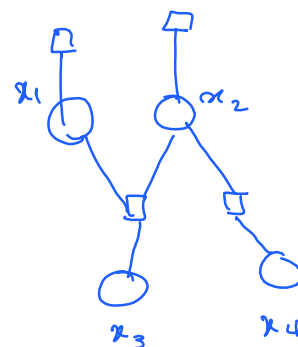
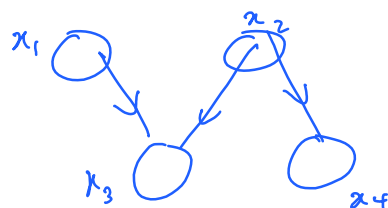
MRF $\rightarrow$ FG

nodes $\rightarrow$ variable nodes
maximal cliques $\rightarrow$ factor nodes



BN $\rightarrow$ FG

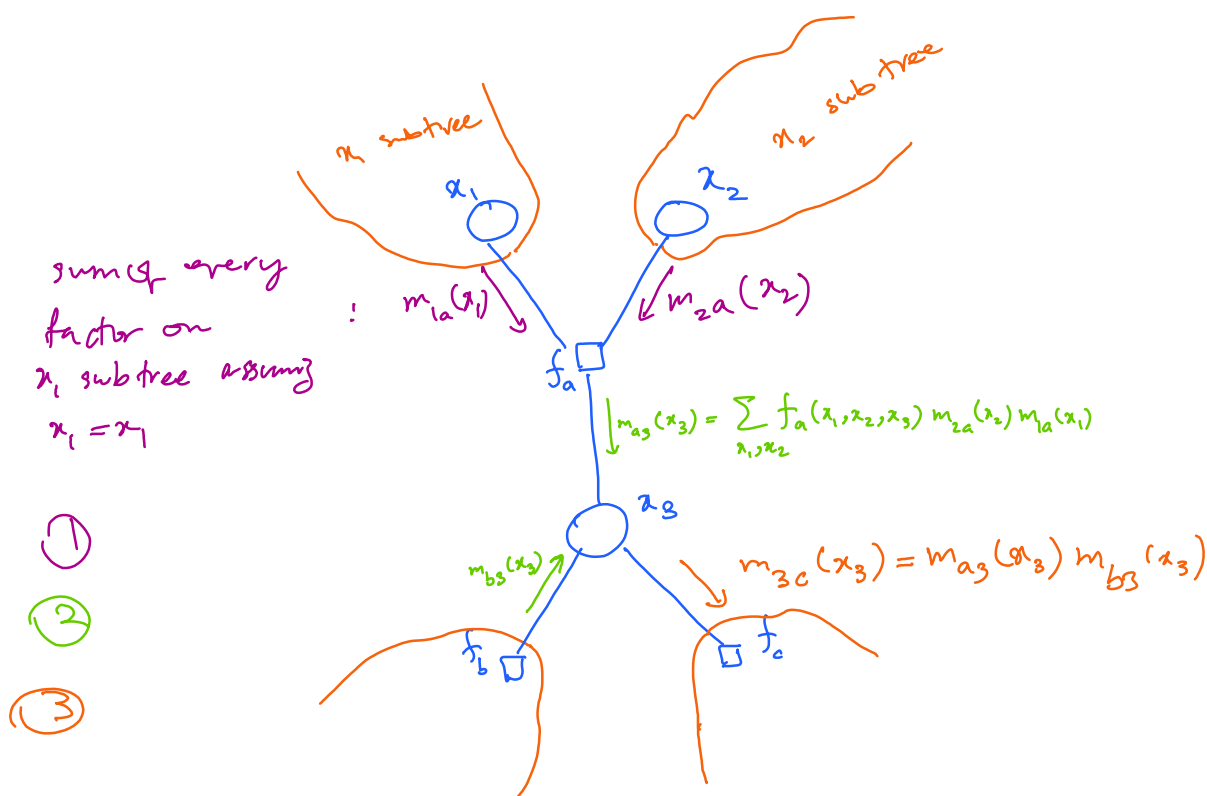
nodes $\rightarrow$ variable nodes
 (CPDs at) nodes $\rightarrow$ factor nodes



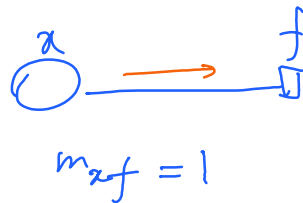
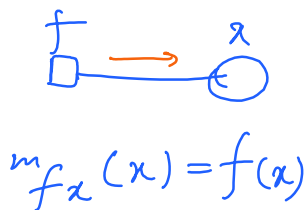
Note: even if the original MRF is not a tree, or the original BN has an MRF which is not a tree, the factor graph may still be a tree: that is discarding the types of the nodes in FG leads to a tree.

this is good because $\rightarrow$

Sum-product for factor tree graphs:



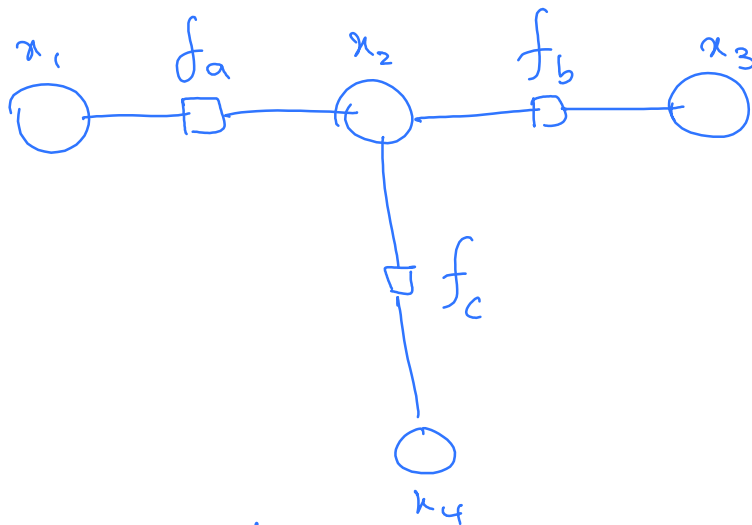
Starting steps at leaves:



Marginal at each node: product of msgs it receives

marginals at all nodes takes twice as much of that of a single node : two msgs per link as opposed to one.

Example:



round 1

round 2

$$\mu_{1a}(x_1) = 1$$

$$\mu_{4c}(x_4) = 1$$

$$\mu_{3b}(x_3) = 1$$

$$\mu_{a2}(x_2) = \sum_{x_1} f_a(x_1, x_2) \mu_{1a}(x_1)$$

$$\mu_{b2}(x_2) = \sum_{x_3} f_b(x_3, x_2) \mu_{3b}(x_3)$$

$$\mu_{c2}(x_2) = \sum_{x_4} f_c(x_2, x_4) \mu_{4c}(x_4)$$

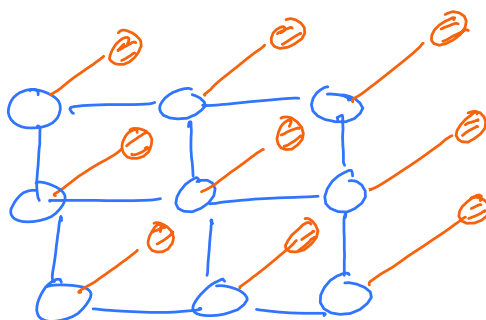
Problem: Identify the most likely configuration:

Find a set of values $x_1^*, \dots, x_m^*$ s.t.

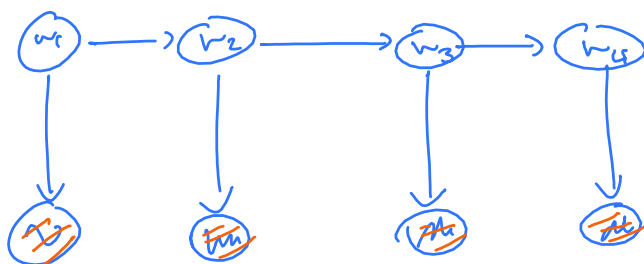
$$P(x_1^*, \dots, x_m^*) \geq P(x_i^m) \text{ for all } x_i^m$$

Applications:

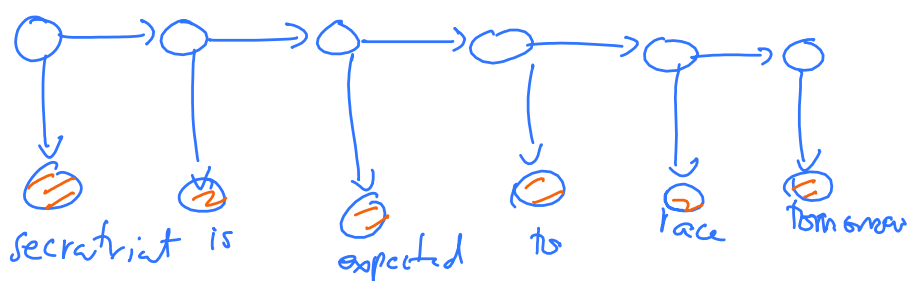
- Image denoising (Lab 1)



- Voice recognition



- Part of speech tagging



Do we already know how to solve this?

Finding most probable state for a node:

run sum-product and find
state with max prob.

Finding most probable configuration for graph:

max-product

$$x_{\max} = \arg \max_x p(x)$$

May not be the same.

	$x=0$	$x=1$
$y=0$	0.3	0.4
$y=1$	0.3	0

$x=0$ max for x

$y=0$ max for y

$(x,y)=(0,1)$ max for (x,y)

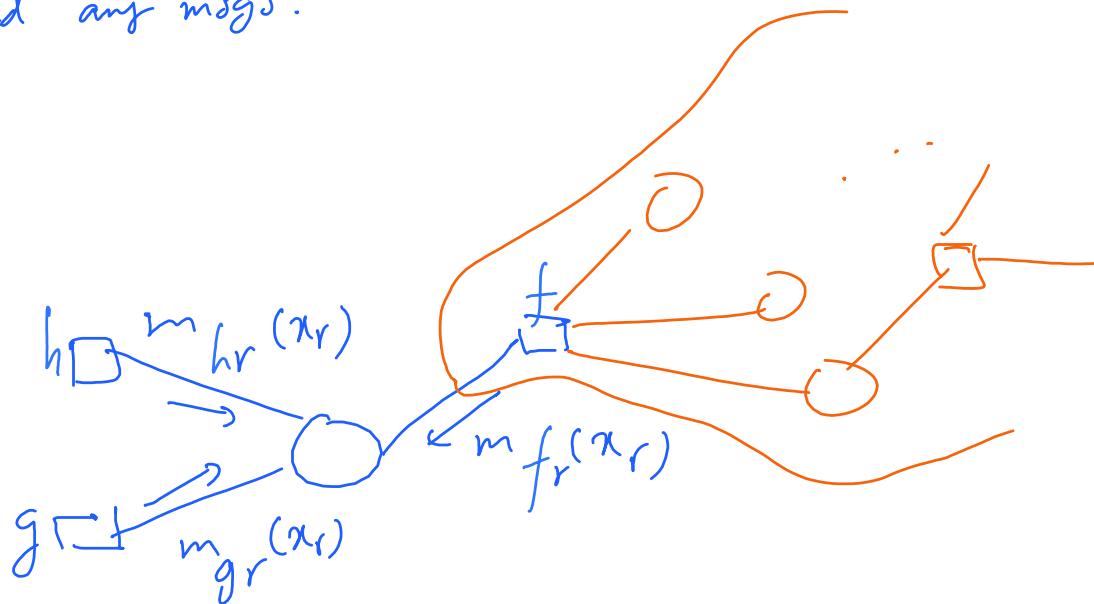
Let's instead try to find $\max_{\vec{x}} p(\vec{x})$

$$\max_{\vec{x}} p(\vec{x}) = \max_{x_1} \max_{x_2} \dots \max_{x_m} p(\vec{x})$$

* We can use a similar approach to elimination
except that Σ is replaced with max.

* Similarly sum-product can be turned
to max-product to find $\max p(x)$

* Here, we pick an arbitrary root, which does not send any msgs.



The message from f $\mu_{fr}(x_r)$

Given the particular value for x_r

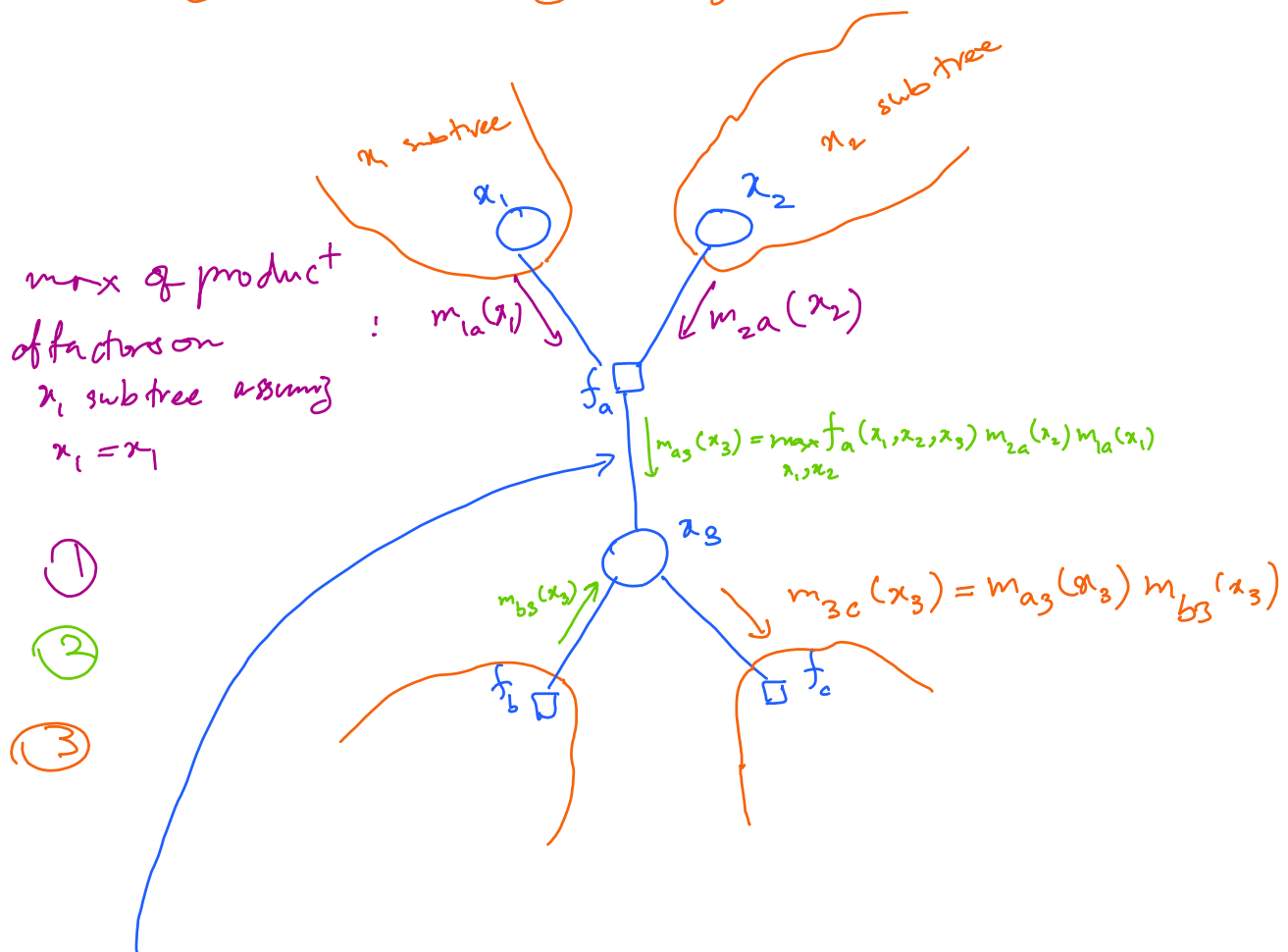
* What is the most likely configuration for the subtree of f

* What is the "probability" of this configuration

$$\max p(\alpha_i^m) = \max_{x_r} \mu_{fr}(x_r) \mu_{gr}(x_r) \mu_{hr}(x_r)$$

from the value of x_r that maximizes this sum and messages we find the most likely configuration

* Finding the maximizing configuration:



For each value of x_3 , we also record which values of x_1, x_2 achieved the max.

At the root, we find x_r^* that achieves the max. Then we back-track and find maximizing values for all nodes.

* It may be more convenient to maximize

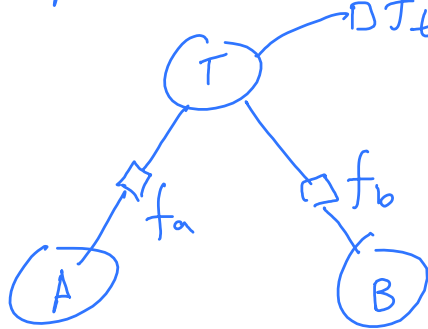
$\ln p(x) \Rightarrow$ max-sum algorithm

$$\sum f_i(x_{f_i})$$

$$\max_y p(x) = \max_{x_1 \dots x_n} \dots$$

note: why not continue the alg so that we can find x_i^* for all nodes just like how we found it for x_r ? We will find maximizing values, but they may belong to different maximizing configurations.

Example: most-likely configuration



$$f_t(T=0) = 0.65$$

$$f_t(T=1) = 0.35$$

$$f_a(A=0, T=0) = 0.9$$

$$f_a(A=0, T=1) = 0.5$$

$$f_a(A=1, T=0) = 0.1$$

$$f_a(A=1, T=1) = 0.5$$

$$f_b(B=0, T=0) = 0.82$$

$$f_b(B=0, T=1) = 0.15$$

$$f_b(B=1, T=0) = 0.18$$

$$f_a(B=1, T=0) = 0.12$$

$$\mu_{A^n}: \max \quad |$$

$$A=0 \longrightarrow |$$

$$A=1 \longrightarrow |$$

$$\mu_{aT} : \max_A f_a(A, T)$$

$$T=0 \longrightarrow 0.9 \text{ for } A=0$$

$$T=1 \longrightarrow 0.5 \text{ for } A=0 \text{ \& } A=1$$

$$\mu_{tT} : \max_T f_t(T)$$

$$T=0 \longrightarrow 0.65$$

$$T=1 \longrightarrow 0.35$$

$$\mu_{Tb} : \max_T \mu_{tT}(T) \mu_{a\bar{1}}(T) f_b(B, T)$$

...